Supplementary material: Coastal-bird responses through Pacific herring spawning in the Strait of Georgia

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1 Scope and version relationship

This supplement accompanies the post-Stage 4A Strait of Georgia event-study refinement. The refinement is additive and post-result. It does not overwrite, relabel, or supersede the registered Stage 4A analysis. The v3 Stage 4A tables, figures, locks, hashes, execution record, and manuscript remain immutable history.

The new complete aggregate family is reported in `outputs/post_stage4a_sog_event_study_v1/`. Manuscript-ready copies are in `tables_v6/`. No protected checklist, observer, locality, source-event identifier, exact coordinate, comment, shoreline field, model object, checkpoint, or record-level derivative is included.

2 Event-study estimand

The refinement asks whether the near/reference difference changes through recorded spawning relative to the same spatial difference during days −28 to −15. Near is <5 km from a recorded source point and reference is 5–20 km. Six joint periods are crossed with the two zones. All concurrent source-event links are counted in their observed joint cell, while one checklist remains one model row.

For period p, the contrast is:

> (near at period p − reference at period p) − (near at baseline − reference at baseline).

Exponentiated detection values are ratios of near/reference odds ratios. Exponentiated conditional-count values are ratios of near/reference geometric-mean positive-count ratios. Values above one indicate a more positive spatial contrast than baseline; they do not estimate population change.

The primary active contrast is the duration-weighted combination of days 0–3 and 4–14. The pre-spawn summary equally combines days −14 to −8 and −7 to −1. All fixed windows and weights were executed without outcome-selected collapse.

3 Complete event-study evidence

The complete family contains 49 core species, both response components where supported, and every registered contrast. `Table_S_event_study_all_49_species_v6.csv` retains estimates, standard errors, confidence intervals, ratios, p-values, BH q-values, model sample sizes, status, and multiplicity family.

The main-species active panel is `Table_3_main_species_active_v6.csv`; its full period companion is `Table_4_main_species_periods_v6.csv`. These files include Hooded Merganser, Surf Scoter, White-winged Scoter, Common Merganser, Glaucous-winged Gull, Mallard, Harlequin Duck, American Crow, Short-billed Gull, Bald Eagle, and Common Raven.

Across estimable core species, BH-adjusted interaction counts were:

Contrast	Detection positive	Detection negative	Count positive	Count negative
Pre-spawn, −14 to −1 d	0	0	1	0
Spawn start, 0–3 d	10	4	13	0
Early egg, 4–14 d	12	2	21	1
Late egg, 15–28 d	10	2	13	0
Primary active, 0–14 d	13	6	19	1

Counts refer to BH $q < 0.05$ within the 49-species outcome-and-contrast family. They are descriptive summaries of correlated contrasts, not independent binomial trials.

4 Specificity comparators

Gadwall and Northern Shoveler were detection-only specificity comparators, not guaranteed negative controls. Their complete results are in `Table_S_specificity_comparators_v6.csv`.

Species	Baseline near/reference	Pre-spawn interaction	Spawn start	Early egg	Late egg	Active 0–14 d
Gadwall	0.88 (0.80–0.97), q = 0.024	0.94, q = 0.419	1.02, q = 0.856	1.03, q = 0.701	1.05, q = 0.506	1.03 (0.90–1.17), q = 0.704
Northern Shoveler	1.00 (0.90–1.11), q = 0.949	1.07, q = 0.419	1.24, q = 0.0917	1.25, q = 0.0114	1.27, q = 0.00246	1.24 (1.08–1.43), q = 0.00564

Gadwall did not reproduce the active interaction. Northern Shoveler did and therefore demonstrates residual time-varying structure. Candidate explanations include zone-specific migration or habitat use, shoreline access, checklist submission, preferential visitation, source-date or source-point classification, and indirect ecology. The comparator result limits causal and trophic-specific interpretation.

5 Model completion and warnings

The family contained 100 species-response fits and 1,400 contrast rows. Model statuses were 95 completed, one completed with a singular warning, three failed for insufficient support, and one failed numerically without fallback.

Species	Outcome	Status	Qualification
Surfbird	conditional positive count	failed insufficient support	joint-cell support gate failed
Rhinoceros Auklet	conditional positive count	failed insufficient support	joint-cell support gate failed
Glaucous Gull	detection	failed numerical fit, no fallback	generic failure code retained; no simplified fit
Glaucous Gull	conditional positive count	failed insufficient support	joint-cell support gate failed
Western Gull	conditional positive count	completed with singular warning	estimate retained with warning

All 22 main-panel fits completed without convergence failure, rank deficiency, or singularity. The smallest released main-panel joint-cell support was 259 rows. Global joint support ranged from 2,992 near-zone checklists at spawn start to 28,655 reference-zone checklists in late egg. Exact year counts for near early- and immediate-pre cells were suppressed because fewer than 20 years were represented.

6 Numerical, multiplicity, privacy, and integrity checks

All 1,344 finite contrast rows reproduced their exponentiated estimates and confidence limits. No written estimate was `NaN` or infinite. BH q-values were independently recalculated across all 42 role-outcome-contrast families and matched with maximum absolute difference (1.78^{-15}).

The production execution used committed code `6bfef2c5b828ca392255d5b3365a2f84b8a2f9f2`. Four protected-input hashes and seven released-output hashes passed. The source-link hash and concurrent-link joint-pairing gates passed. Zero records from 2026 onward were read.

The release privacy scan detected no checklist, observer, locality, coordinate, credential, or protected-path violation. It emitted known invalid-UTF-8 read warnings while traversing non-UTF-8 artifacts; no violation resulted. Frozen Stage 4A outputs and decision records had no tracked changes.

7 Historical registered Stage 4A analyses

The following registered Stage 4A outputs remain part of the complete scientific record and are retained unchanged:

- historical M05 timing and distance family: `tables_v3/Table_S2_all_M05_time_distance_effects_v3.csv`;
- historical M05 guild timing family: `tables_v3/Table_S2a_guild_event_time_effects_v3.csv`;
- historical M08 active-minus-contemporaneous-reference contrasts: `tables_v3/Table_S12_M08_active_minus_reference_v3.csv`;
- historical M02 complete species effects: `tables_v3/Table_S1_all_species_effects_v3.csv`; and
- historical M29 specificity results: `tables_v3/Table_4_specificity_sensitivity_v3.csv`.

M05 used the registered six-window timing structure with early pre-spawn (−42 to −29 d) as baseline and separate additive time and distance margins. M08 was the registered active-minus-reference contrast for the historical active classification. Neither is the present joint period-by-zone difference-in-differences. The historical legacy aggregate release also did not identify the component-level engine after its allowed fitting path. Those facts are preserved rather than repaired retrospectively.

8 Reproducibility and governance

Fixture execution tested exact period and zone boundaries, concurrent-link pairing, join cardinality, detection/count semantics, and protected-input guards. Production required the explicit acknowledgement `through_2025_post_result_refinement_v1` . No result was used to change the fixed periods, primary 0–14 day weight, species definitions, eligibility, model structure, or specificity panel.

Current results are exploratory and estimand-refining until prospective confirmation. The 2026–2028 response remains locked and may not be inspected through an interim look. A future confirmation must use the complete prospective release and an approved execution package.

9 Supplementary file inventory

File	Content
`Table_3_main_species_active_v6.csv`	11-species primary 0–14 day interactions
`Table_4_main_species_periods_v6.csv`	baseline, pre-spawn, onset, egg, and active contrasts for 11 species
`Table_S_event_study_all_49_species_v6.csv`	complete core-species effect family
`Table_S_specificity_comparators_v6.csv`	Gadwall and Northern Shoveler detection families
`model_diagnostics_v1.csv`	all fit states, warnings, and failures
`model_term_support_v1.csv`	joint-cell model support
`joint_exposure_support_v1.csv`	global period-by-zone support
`execution_record_v1.yml`	authorization, hashes, counts, and final gate
`output_hash_manifest_v1.csv`	hashes for released aggregate outputs